

Study on friction stir diffusion bonding of aluminum to zinc-coated steel: A comparison to weld-brazing

K. Torabi^a, R. Beygi^{a,*}, A. Alikhani^a, E.A.S. Marques^b, A. Khalfallah^{c,d,**}, L.F.M. da Silva^b

^a Department of Materials Engineering and Metallurgy, Faculty of Engineering, Arak University, Arak 3815688349, Iran

^b Department of Mechanical Engineering, Faculty of Engineering of the University of Porto, Rua Dr. Roberto Frias, Porto 4200-465, Portugal

^c CEMMPRE, Department of Mechanical Engineering, University of Coimbra, Rua Luís Reis Santos, Coimbra 3030-788, Portugal

^d Laboratoire de Génie Mécanique, École Nationale d'Ingénieurs de Monastir, Université de Monastir, Av. Ibn El-Jazzar, Monastir 5019, Tunisia

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ABSTRACT

This paper investigates the efficacy of Friction Stir Diffusion Bonding (FSDB) for joining aluminum (AA 1050) to zinc-coated steel (galvanized carbon steel) in a lap-joint configuration. The process was conducted with a tool rotation speed of 950 rpm, a welding speed of 20 mm/min, and a pin penetration depth of 2.8 mm, ensuring the pin remained above the steel surface. FSDB, a variant of Friction Stir Welding (FSW), differs primarily in its non-penetrative approach to harder materials, particularly steel. This unique FSDB approach minimizes tool wear and reduces thermal stresses. The zinc coating on steel plays a crucial role in promoting metallurgical bonding with aluminum through the stirring action of the tool. Scanning Electron Microscopy (SEM) and energy-dispersive X-ray Spectroscopy (EDS) analyses of the interface layer revealed a continuous Al-Fe-Zn solid solution, which contrasts with Al-Fe-Si intermetallic layers (IMCs) typically observed in Al/St joints produced via Weld-Brazing (WB). The reduced heat input in the FSDB, which can be attributed to the lack of tool penetration into the steel, prevented the evaporation of zinc and thus minimized the porosity within the joint. Mechanical testing, including tensile shear strength and microhardness assessments. The heat-affected zone (HAZ) and stir zone (SZ) exhibit hardness values ranging from 44.9 to 55.5 HV, indicating that FSDB joints exhibit superior mechanical properties and less brittleness than WB-fabricated joints. FSDB exhibited superior joint quality (3830 N fracture load for 10 mm width), less brittleness (fracture from Al and not from the interface), elimination of IMCs layer, minimized porosity, and controlled thermal input making it a promising technique for dissimilar metal joining in lightweight applications.

1. Introduction

The widespread use of lightweight aluminum alloys in industrial applications has led to increased attention and efforts toward the dissimilar joining of lightweight metals such as aluminum to magnesium (Al-Mg) and aluminum to steel (Al-steel) [1]. Notably, the joint of Al-steel offers the combination of advantages and physical properties of individual materials, broadening its application [2,3]. Despite these benefits, joining Al-steel is challenging due to the large difference in melting points, low solubility of iron in Al, and formation of brittle intermetallic compounds (IMCs) at the interface, which can adversely affect the mechanical properties and reliability of joints [4]. The formation of IMCs layers promotes sufficient bonding at the Al-steel

interface, thereby preventing potential fracture [5]. In this regard, to benefit from the advantages of IMCs, the thickness of the IMCs layer should be adjusted to within a critical limit of $\sim 10 \mu\text{m}$ [6]. Furthermore, at the interface of Al-steel, the presence of IMCs layer composed of Fe_xAl_y confirms effective binding between Al and steel [7]. On the other hand, Al-rich IMCs (e.g. FeAl_3 , and Fe_2Al_5) cause brittle joint deteriorating mechanical properties [8,9]. Variations in the carbon percentage in steel lead to different mechanisms in Al-steel joints because carbon controls the solidification process as well as the shape and thickness of the IMCs layer [10]. On the other hand, IMCs layer composition at the stir zone depends on the engaged welding processing parameters [11]. Therefore, improving the mechanical properties is linked to practical joining parameters such as tool hardness, heat input, traverse speed, and

* Corresponding author.

** Corresponding author at: CEMMPRE, Department of Mechanical Engineering, University of Coimbra, Rua Luís Reis Santos, Coimbra 3030-788, Portugal.

E-mail addresses: r-beygi@araku.ac.ir (R. Beygi), ali.khalfallah@dem.uc.pt (A. Khalfallah).

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rotating speed, which can lead to the development of effective and efficient methods for dissimilar joining of aluminum to steel [12,13].

To maximize the mechanical properties of Al-steel joints, an optimized heat input can be used to control the formation and thickness of IMCs layer [14]. There is a plenty of weld- processes such as cold metal transfer (CMT) [15], friction stir welding (FSW)[16], diffusion bonding [17], friction stir spot welding (FSSW) [18], laser welding/brazing (LWB) [19], friction stir spot welding-brazing (FSSW-B) [20], and TIG, welding-brazing (WB), which differ in terms of equipment, parameters, mechanism, and performance [10]. An appropriate dissimilar joint between Al and steel can be achieved through a combination of FSW and diffusion bonding known as friction stir diffusion bonding (FSDB) [21]. This process exploits the frictional heat and mechanical pressure produced by FSDB to diffuse atoms at the interface between materials. In other words, the weld tool generates appropriate heat to cause atomic diffusion without reaching the melting point of the base materials [22]. Because of atomic diffusion, an IMCs layer is formed at the interface, affirming metallurgical bonding. The FSDB configuration process is almost similar to FSW without pin penetration in the retarding sheet maintaining the thermal-mechanically affected zone (TMAZ) above the lap interface and covering the heat-affected zone (HAZ) simultaneously [23]. In the FSDB process, maintaining the pin penetration depth above the steel surface is crucial to minimize direct contact between the tool and the steel. This approach prevents excessive mechanical mixing, which could otherwise increase the thickness of IMCs, and preserves the zinc layer as a diffusion barrier, thereby controlling atomic interactions at the interface [23,24]. Slight pin penetration into the aluminum layer, without contacting the steel, optimizes joint strength while avoiding the formation of brittle IMCs [24]. When the pin penetration depth is optimized, it ensures sufficient stirring and bonding at the interface without excessive heat input that could lead to the formation of large amounts of IMCs[25]. As a result, atomic diffusion occurs due to the large forging force produced on the interface by the tool shoulder [26]. As previously mentioned, FSDB is a welding method that does not involve pin penetration, which offers several advantages. Notably, FSDB has been shown to minimize tool wear by preventing direct interaction between the tool and steel, thereby reducing mechanical stress and contact on the tool [16,23,27]. This method significantly reduces tool wear compared to FSW, where the tool frequently engages with harder materials, leading to increased tool degradation. Furthermore, FSDB hinders the formation of defects, hooks, distortion stresses, and cracking because of the absence of a liquid phase at the interface [28,29]. However, it has been reported that in friction stir lap welding, double-pass welding is required to remove defects and IMCs layer [30]. It should be considered that to maximize joint quality, the FSDB parameters (e.g., tool rotation speed, dwell time, and pressure) should be precisely optimized. Otherwise, incomplete bonding or excessive IMCs and material flow at the interface can drop the joint quality [27]. Furthermore, metal sheets are prone to deformation and distortion due to the high temperatures and pressures, resulting in decreased joint quality [22]. To address these challenges, various interlayers have been investigated to enhance joint quality. Nickel, titanium, and copper-based interlayers are commonly used for their ability to mitigate IMC formation and improve mechanical properties through diffusion bonding [5–7]. Nickel interlayers, for example, promote the formation of ductile phases, while titanium interlayers enhance bonding strength by forming stable titanium-aluminum phases [8]. Although brass interlayers can improve the metallurgical bond between aluminum and steel, it presents several disadvantages, such as lower joint strength, less effective flux, and complex interfacial reactions [31]. Copper-based interlayers can improve thermal conductivity at the interface but often lead to thicker IMC layers, which can compromise joint reliability [9]. As the zinc coatings on steel (known as galvanized steel) act as a thermal barrier, reducing heat transfer to the steel, the zinc interlayer offers unique advantages for Al-Steel joints [32]. Notably, zinc acts as a lubricant during the process in a way that zinc diffuses into the

Table 1

Chemical compositions (wt%) of base AA 1050 and St 37 in as received conditions.

Element	Chemical composition		
	AA 1050	St 37	Galvanized zinc
Al	99.5	-	0.0001
Fe	< 0.4	99.76	0.002
Zn	< 0.05	-	99.97
Si	< 0.25	< 0.002	-
Mg	< 0.05	-	0.02
Cr	-	< 0.012	-
Mn	< 0.05	< 0.224	-
Ti	< 0.03	-	-
C	-	< 0.15	-
Cu	< 0.05	-	0.002

aluminum sheet, resulting in the formation of a large bonding area and strength of joints [33,34]. Furthermore, zinc coating provides a thermal barrier between aluminum and steel, thereby accelerating heat loss due to zinc's thermal diffusivity, which is approximately three times that of bare steel [35]. As a result, while IMCs layer formation is reduced, the wettability and compatibility between Al and steel are improved, resulting in improved shear tensile strength in the joint zone [36,37]. Through the process of zinc diffusion into aluminum, a ternary solid solution is formed. This diffusion process significantly enhances the bonding strength between the materials, leading to greater durability and cohesion. [32,33]. In contrast to weld-brazing, in which the zinc layer evaporates during the process leading to porosity in the joint zone and reduced joint strength [20], FSDB limits the temperature increase, thus maintaining the zinc layer at the interface. As a result, the formation of porosity and hooks is restricted at the interface. To improve the mechanical properties of weld joint, researchers are striving to optimize the parameters involved. For instance, Singh et al. [38] optimized the gap between materials to accelerate the escape of zinc fumes, resulting in a 67–80 % increase in joint efficiency. It is well documented that welding speed can affect the tensile properties and fracture locations of joints in friction stir lap welding of aluminum and low-carbon galvanized steel [39,40]. The pin penetration depth into the steel sheet was considered a dominant parameter in the weld strength; researchers observed higher joint strength when the probe pin reaches the steel surface [41,42]. To this end, tool materials should be formed from materials with higher strength than steel, such as carbide, which is not cost-efficient. Ji et al. [23] proposed positioning the pin on a steel surface such that the bottom of the pin is close to the steel surface without penetrating it. As another prime example, Haghshenas et al. [24] used the FSDB approach to join dissimilar materials, namely 5754 aluminum alloy and coated high-strength steels (DP600 and 22MnB5), with a pin positioned 0.10 and 0.05 mm above the steel sheet. The results confirmed that the presence of Al-Fe composition at the interface lead to strong joints and less brittle fracture.

While conventional methods like weld brazing encounter challenges with porosity due to zinc evaporation, and friction stir welding faces issues with hook formation and material defects, this work introduces a unique approach using FSDB. The innovative application of FSDB to the aluminum/galvanized steel system, with its non-penetrative nature minimizing tool wear and thermal stresses, is specifically aimed at achieving defect-free joints. Furthermore, this study focuses on the key role of zinc coatings in facilitating metallurgical bonding. The lower heat input in FSDB minimizes zinc evaporation, thereby not only reducing porosity but also controlling IMC formation. The mechanisms by which zinc promotes atomic diffusion and how this affects IMC formation and resulting joint quality are explored and discussed. Ultimately, this research elucidates the parameters governing FSDB, with the aim of understanding the fundamental mechanisms contributing to the formation of high-quality, defect-free joints in dissimilar material combinations.

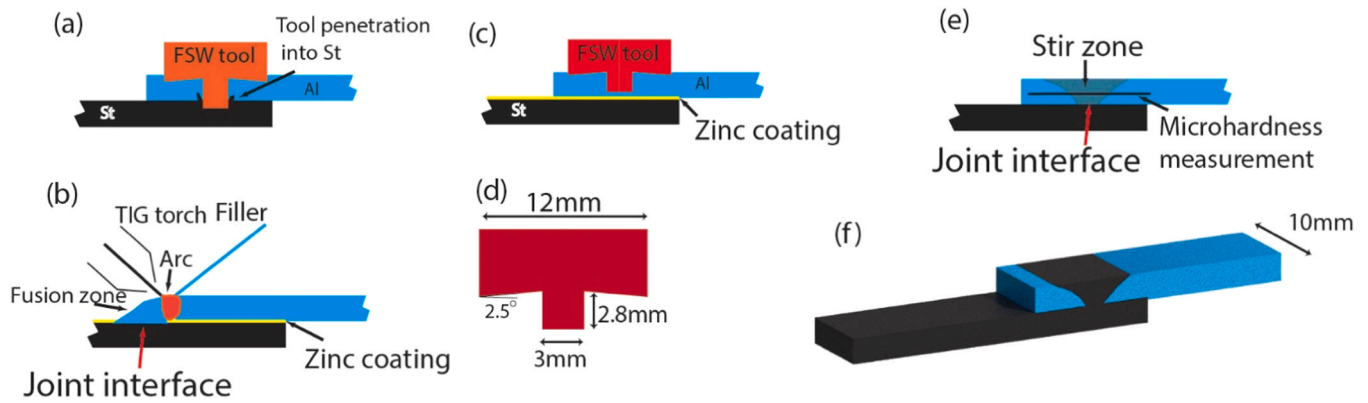


Fig. 1. Schematic representation of joining configurations: (a) conventional FSW, (b) WB technique, (c) and FSDB process. (d) dimensions of the FSW tool. (e) Microhardness measurement line. (f) specimen for shear tensile testing.

As a part of our ongoing projects on Al to galvanized steel welding, here, the present results are compared with the obtained results of the WB method published elsewhere [43].

2. Materials and methods

In the present work, AA 1050 (thickness: 3 mm, in rolled condition) and galvanized steel (ST 37, thickness: 5 mm, low carbon) were used as base sheets in but-joint FSDB. The chemical compositions of the received base metals are listed in Table 1. To remove oil and oxide films from the surfaces, before welding, the pieces were cleaned with acetone followed by brushing with a wire brush. In conventional FSW, the pin is penetrated into St, causing detachment of St and local sever mixing of St and Al, as shown in Fig. 1a. In WB a heat source is used to fuse and speared both Al and filler material over the surface of steel, as shown in Fig. 1b. In the present study, the FSW tool stays away from the surface of steel, as shown in Fig. 1c. FSDB was performed under the following conditions: rotation speed of 950 rpm, welding speed of 20 mm/min, tilting angle of the tool of 2.5°, and the pin penetration depth of 2.8 mm. The comparison of Al thickness and pin penetration depth revealed that pin remained above the steel-sheet surface. For the metallurgical analyses, joints were sectioned perpendicularly to the welding direction followed by polishing. The geometry of the applied tool to the FSDB process is shown in Fig. 1d. The microstructural analysis was carried out using a scanning electron microscope (SEM) equipped with an energy-dispersive X-ray (EDX) spectroscopy system. SEM images were taken in back scatter (BS) mode which makes a chemical composition contrast, helping to distinguish the IMCs. Microstructural characterizations by SEM have been performed on the joint interfaces shown in Fig. 1b and e for WB and FSDB joints, respectively. The elemental distribution at the interface was also studied by elemental mapping analysis. The mechanical features were explored by shear testing in an alignment in which spacers were used to hinder the bending and rotation of the joint during the test. The microhardness of metallic materials was determined using the Vickers hardness test method in which a load was imposed on the specimen to measure its hardness, where a load of 100 gf was applied for 15 s. The microhardness test was performed along a line starting from Al, across the welding zone, to the Al, as shown in Fig. 1e. Tensile specimens with 10 mm width were cut as shown in Fig. 1f. Shear tensile strengths were recorded for 3 specimens. Tensile testing was performed at the cross-head speed of 2 mm/min.

3. Results and discussion

3.1. Microstrural analyses

To visualize the microstructural features of the weld-zone, SEM analysis was performed in an area in which welding interface can be

explored. Fig. 2a and b show SEM images taken at different magnifications. The images revealed that the two sheets were tightly bonded without indentation. As can be seen in Fig. 2b, at the Al–steel interface, the bonding morphology is almost flat without hooks and cavities, confirming that FSDB managed to eliminate downsides related to the FSW method. This phenomenon is probably due to the presence of a zinc layer, which remarkably inhibits heat flow into the steel sheet. To track the elemental distribution, the chemical compositions of points 1, 2, and 3 were determined by spot EDS; these results are shown in Fig. 2b, and their results are listed in Table 1. Notably, it is seen that points 1 and 3 are close to AA 1050 and St37, respectively, and point 2 is located in the middle of the transition layer. Fig. 2c shows the EDS results. At all points, the presence of Al, Fe, and Zn was confirmed by the observed picks. For point 2, according to the welding procedure, the presence of Al and Fe in this region is probably due to the imposed heat and pressure resulting in Al and Fe diffusion. The presence of Zn could be explained by the low heat input at the zinc melting and evaporation point. Notably, the Al:Fe atom ratio is approximately 3:1, which could be associated with the formation of FeAl_3 IMCs due to effective heat input [25]. On the other hand, the presence of Al and Fe could be related to slight diffusion in the Zn layer. Contrary to this, at points 1 and 3, the percentages of Al and Fe is higher than the rest, which could be ascribed to the diffusing of these elements in the Zn layer due to heat input. As a supplementary investigation, line scan analysis was employed to monitor elemental changes across the transition layer (i.e., Al-side to the steel side), the scan region and results are presented in Fig. 2d. The elemental spectrum depicted in Fig. 2e, suggested that by moving across the interface, the Al content significantly decreases at the bonding interface; while, Fe and Zn percentages increase. In particular, the presence of Al and Fe in the bonding region indicates that Al and Fe are diffuse into the Zn layer. Note that the IMCs thickness is under 10 μm which benefits the bonding strength. The lower percentage of Fe in the Zn layer was because the pin bottom did not infiltrate into the steel layer, as shown in the welding configuration procedure. Therefore, it can be concluded that the mixing layer contained either FeAl_3 or a ternary solid solution of $\text{Al}_{71.5}(\text{Fe}_{19}, \text{Zn}_{9.5})$. The formation of a solid solution, specifically $\text{Al}_{71.5}(\text{Fe}_{19}, \text{Zn}_{9.5})$, significantly enhances joint strength compared to IMCs like FeAl_3 . This improvement is attributed to the uniform distribution of aluminum, iron, and zinc atoms within the crystal lattice, which reduces localized stress concentrations that can lead to cracks in IMCs. Additionally, the solid solution exhibits increased ductility and reduced brittleness, which contributes to its superior mechanical properties. The lattice distortions caused by the solute atoms impede dislocation movement, thereby increasing the material's yield strength. Overall, the ductility and toughness of the solid solution, such as $\text{Al}_{71.5}(\text{Fe}_{19}, \text{Zn}_{9.5})$, make it more effective in enhancing joint strength compared to the brittle nature of IMC phases [7,8,44,45]. Furthermore, the formation of this solid solution indicates a homogeneous distribution of aluminum, iron, and zinc

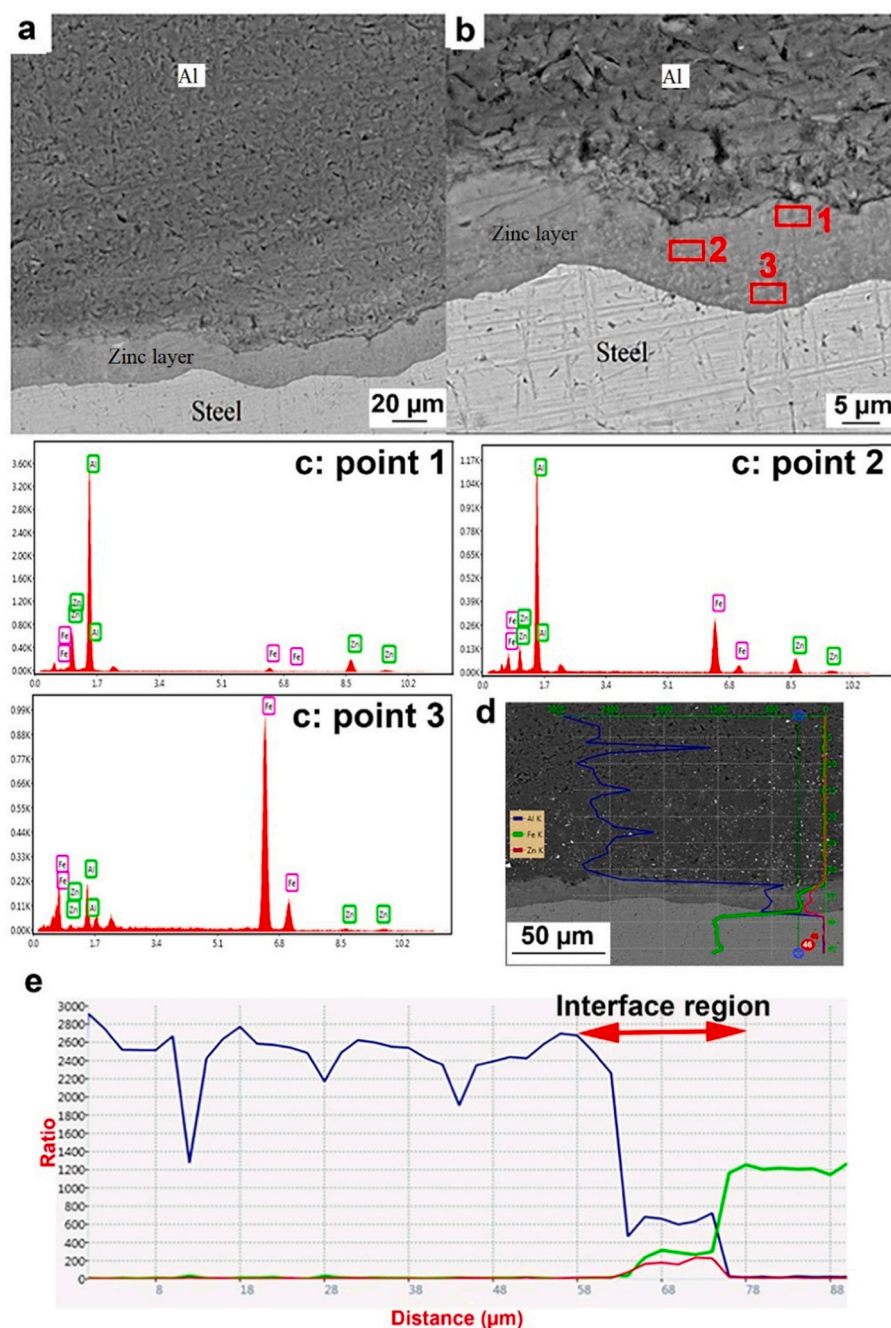


Fig. 2. (a and b) SEM images of the fabricated sample at different magnifications, (c) EDS plane scan analysis at points 1, 2, and 3, and (d and e) EDS line scan analysis of elements across the interface.

Table 2

Element content (at%) and possible phase at mentioned points in Fig. 2b.

Point	Al	Fe	Zn	Possible phase
1	87.60	1.80	11.14	Solid solution
2	71.60	1.88	9.53	FeAl ₃ or solid solution of Al _{71.5} (Fe ₁₀ Zn _{9.5})
3	21.38	76.99	1.64	Solid solution

atoms, which helps fill potential voids and ensures a continuous, defect-free bond layer, thereby preventing porosity.

Compared to reported methods [46,47], the proposed method eliminates formation cavities and hooks at the interface and requires no further modification. For instance, Li et al. [48] recommended that using a pinless tool and applying subsequent tool stirring processes led to

free-defect bonding. In another report, the use of pinless tools with an l-shaped groove on the shoulder with a large plunge depth reduced hook and cavities formation [47]. However, in the present work, the pin maintained above the steel surface, and the pin penetration in the steel was zero. Table 2

To assess the possibility of formation of IMCs at the interface, the liquidus projection line of the ternary Al-Zn-Fe system, which represents the boundary between the liquid and solid phases in a three-dimensional phase diagram, is shown in Fig. 3. This diagram illustrates the different segments and regions describing variant reactions and phases in a ternary system. As can be seen, the liquidus projection line of the Al-Zn-Fe system is located in the Al corner, indicating that the sample has a high Al content and low Zn and Fe content. As documented by others [44], the formation of FeAl₃ is suppressed by the presence of a certain

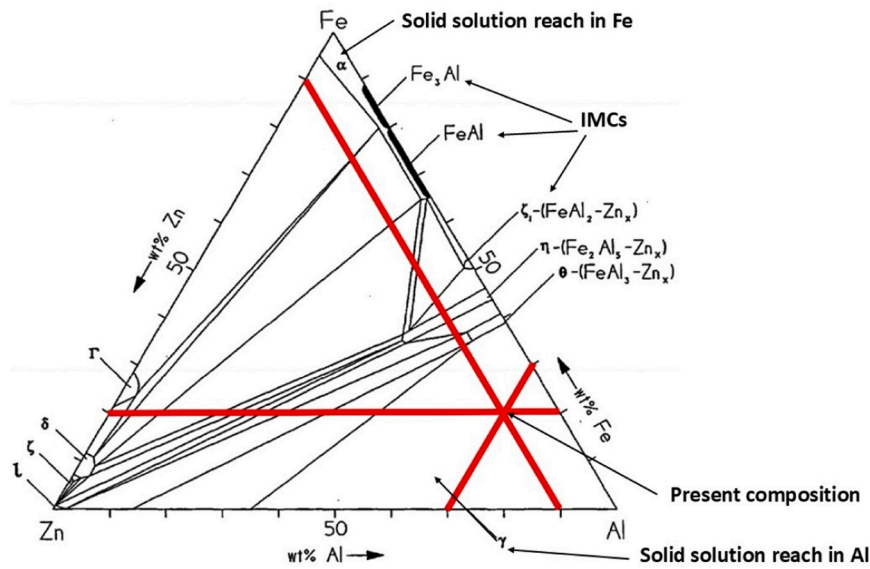


Fig. 3. Liquidus projection lines of the ternary Al-Zn-Fe system. The red line represents the composition of the interface layer. Adapted from Ref. [49].

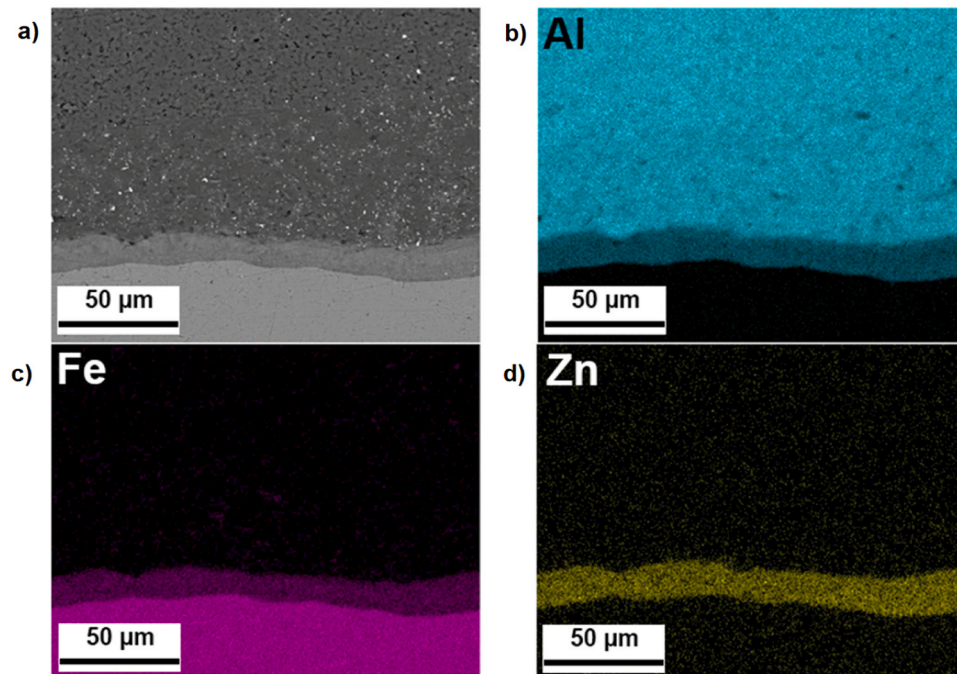


Fig. 4. Elemental mapping of the atomic distribution at the Al-Zn-coated steel interface of the prepared sample: (a) Backscattered electron (BSE) image of the cross-sectional interface, showing the overall microstructure; (b) distribution of aluminum (Al) atoms, highlighting the aluminum-rich region; (c) distribution of iron (Fe) atoms, indicating the presence of the steel; (d) distribution of zinc (Zn) atoms, showing the diffusion layer at the interface. Scale bars represent 50 μm .

amount of Zn. Furthermore, another study showed that stable FeAl_3 can be formed in a narrow region near the Fe-Zn binary edge [45].

Therefore, according to the EDS results and the liquidus projection diagram, the presence of Al and Fe in the Zn layer is due to low atomic diffusion as well as low mixing of Fe due to the absence of contact between the tool and steel, which results in the formation of a solid solution. It is predicted that the absence of an IMCs layer improves the strength by rendering it less brittle.

Further investigation for the elemental distribution of the samples was performed by elemental mapping, as shown in Fig. 4. The presence of Al, Fe, and Zn was confirmed, which is consistent with the EDS plane and line analysis. As can be seen, the penetration of Al into Fe and vice

versa is hindered by the presence of the Zn layer.

Considering reports of Al and galvanized steel welding by WB [43], at the interface different IMCs layers were formed which is dependent on the filler material. It is documented that the IMCs layers compositions strongly rely on the filler material's content but not the Zn layer. On the other hand, Zn evaporation in WB creates pores at the interface. However, using the FSDB approach limits the formation of IMCs and pores at the welding zone.

3.2. Microstructural analyses

The microhardness test was carried out on the Al side along with a

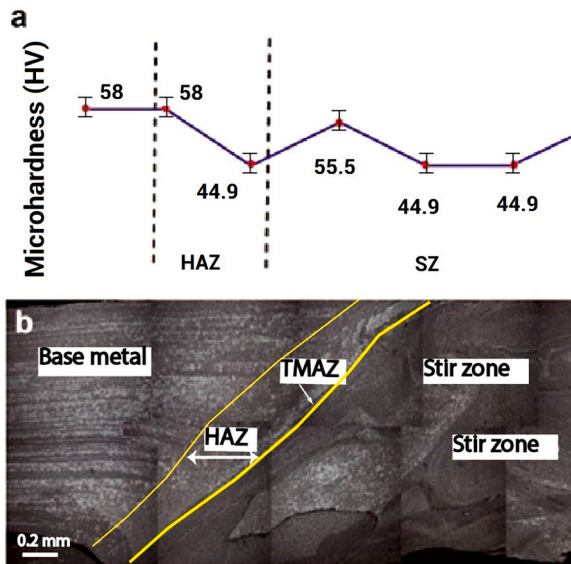


Fig. 5. (a) Hardness profile of the joint and (b) SEM images of the aluminum region and identification of different areas, such as the base metal (BM), heat-affected zone (HAZ), thermos-mechanically affected zone (TMAZ), and stirring zone (SZ).

line parallel to the interface across the AA 1050-HAZ-SZ-HAZ-AA 1050 region. The microhardness results are shown in Fig. 5a. For AA 1050, the microhardness was measured at 60 HV. Although this result is not a very

high number, considering the absence of a hard phase and precipitate in the AA 1050 sheet, this hardness level is acceptable. The HAZ zone possessed the lowest hardness level, indicating that input heat caused softening. Particularly, the microhardness fluctuated in the SZ region, which could be explained by different heat input and solidification conditions. The influence of grain size on hardness level was investigated using SEM images taken from different areas. Fig. 5b shows that the grain size in different regions is completely non-uniform and is mostly influenced by the pin movement on the advancing side. The base metal zone (BM) had a grain size dependent on the rolling direction, which as shown in the image, the grain orientation was completely horizontal. The disparity in temperatures between the AS and RS results in distinct variations in hardness values within the HAZ. Furthermore, the HAZ area is very small and a slightly coarse grain was observed in this area. In the SZ region where the pin tool applied pressure and heat, the grain size was diverse and increased. In the thermo-mechanically affected zone (TMAZ), which narrowly overlaps the HAZ and SZ regions, changes in the microstructure are noticeable. Considering the above-mentioned results, in general, a smaller grain size tends to increase the microhardness level. But, on the other hand, the dynamic recrystallization in the stir zone causes elimination of dislocations. As the base metal of Al is in work-hardened condition, a slight drop in hardness of the stir zone from 58 to 44.9 is observed due to this dynamic recrystallization.

The elongation was 7.5 %, mainly due to the desirable connection of the AA 1050 to St37. The ultimate tensile strength (UTS) of the fabricated sample was approximately 3830 N. The high tensile strength is owing to the fact that during the welding process, the heat input is reduced by the zinc layer and IMCs do not form at the interface, which

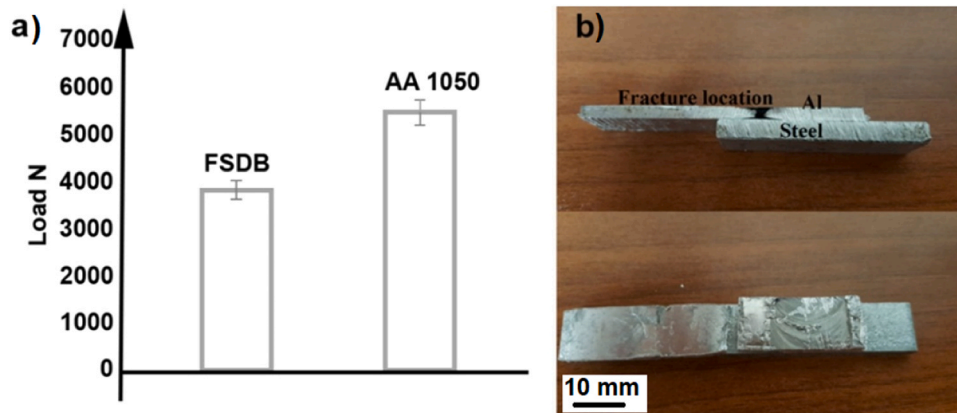


Fig. 6. (a) Examined shear load-displacement diagram of the shear tensile tests on the welded sample and base metal and (b) the fracture location.

Table 3

Comparison of the tensile test results of the FSDB and WB methods.

M1	M2	Technique	Welding Parameters	IMC layer	UTS (N)	Ref.
AA1050	Carbon steel	WB	Welding speed: 10 mm/min Gas flow rate: 15 L/min welding current: 70 A Filler: 4043	$\text{Fe}_4(\text{Al},\text{Si})_{13}$ $\text{Fe}_2\text{Al}_8\text{Si}(\tau_5)$	3359 ± 130	[42]
AA1050	Carbon steel	WB	Welding speed: 10 mm/min Gas flow rate: 15 L/min welding current: 70 A Filler: 4047	$\text{Fe}_2\text{Al}_8\text{Si}(\tau_5)$ $\text{Fe}_2\text{Al}_9\text{Si}_2(\tau_6)$	3257 ± 90	[42]
AA1050	St 37 with zinc coating	FSDB	rotational speed: 950 rpm Welding speed: 20 mm/min Pin penetration: 2.8 mm	Solid solution of $\text{Al}_{71.5}(\text{Fe}_{19},\text{Zn}_{9.5})$	3830	This work

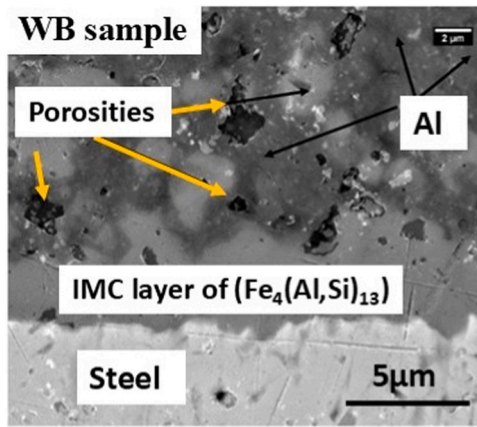


Fig. 7. Cross section of aluminum-steel joint made by WB showing porosities in the matrix of Al and IMCs layer at the interface.

could significantly improve the joint strength. Fig. 6b shows that the fracture was located in the HAZ area which is the softened region. It can be observed that the fracture mode is ductile.

A comparison of the results from FSDB and previous findings obtained by WB [42] method reveals that the tensile shear strength was remarkably increased, (see Table 3). This result can be explained by the presence of Zn and a low amount of Fe causes a solid solution to form at the interface. The solid solution phases possess higher fracture toughness than IMCs because of the presence of metallic bonds instead of covalent bonds [50,51]. Indeed, the zinc layer effectively regulates the diffusion of aluminum and iron atoms, resulting in the formation of a ternary solid solution such as $Al_{71.5}(Fe_{19}, Zn_{9.5})$ [32,33], as confirmed by EDS analysis. This controlled diffusion process helps to mitigate the formation of brittle phases like $FeAl_3$ and Fe_2Al_5 , which are commonly observed in other techniques. The comparison of the results shows that the mechanical properties are strongly correlated to the formation of possible phases, suggesting that a joint without brittle IMCs phases at the interface results in improved mechanical properties. Furthermore, the FSDBs enhanced the joint strength by eliminating brittle IMCs (i.e., $FeAl_3$, Fe_2Al_5 , $FeAl_2$), providing Fe-rich compounds (i.e., $FeAl$), and creating a solid solution of Fe in Al, which tends to increase the interface strength in Al/St joints. Further details on phase analyses for WB techniques are provided in reference [43], allowing to compare the phase evolution in two techniques in more details. As an example, Fig. 7 shows the interface of a WB joint. The presence of IMC layer as well as porosities caused by Zn evaporation are shown in this figure.

It can be concluded that two factors affect the joint strength of the aluminum-galvanized carbon: mixing pattern and microstructure features. The formation of hooks/cavities and hot-cracking negatively affect the load-bearing capacity, increasing the stress concentration, and weakening the joint. In this study, no hook is observed in the joint, indicating that no weak interface is present perpendicular to the loading direction. Furthermore, pores are absent at the interface, which occurs during weld brazing.

4. Conclusions

This study demonstrates the effectiveness of FSDB for joining dissimilar materials, aluminum (AA 1050) to galvanized carbon steel, in a butt-joint configuration.

The key findings of the research are as follows:

1. The tensile shear strength of the joint reached 3830 N, demonstrating a strong bond.

2. Microstructural analysis revealed the formation of a thin, continuous IMC layer with a thickness of less than 10 μm , indicating controlled diffusion and minimal formation of brittle phases.
3. The FSDB process successfully eliminated common defects associated with dissimilar joining techniques, such as hooks and porosity, leading to a homogeneous and defect-free interface.
4. Compared to conventional weld-brazing, FSDB offers significant improvements. The process operates at lower temperatures, reducing the risk of excessive IMC growth and the evaporation of zinc, which often compromises joint quality in weld-brazing. This results in enhanced joint strength and superior microstructural characteristics.
5. The innovative use of the FSDB technique emphasizes its potential as a viable and sustainable method for dissimilar metal joining, particularly for lightweight applications. This research is a promising approach, as it is the groundwork for future research on optimizing FSDB parameters and broadening its use in other dissimilar material systems.

CRedit authorship contribution statement

Marques EAS: Methodology, Formal analysis. **Alikhani A.:** Investigation, Data curation. **Beygi R.:** Writing – original draft, Supervision, Data curation, Conceptualization. **Torabi K.:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **da Silva LFM:** Resources. **Khalfallah Ali:** Writing – review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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